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Cancer Vaccine Development: Toward Artificial Intelligence-Assisted Personalized Cell Membrane Nanovaccine

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ABSTRACT

Cancer vaccines are an attractive immunotherapeutic approach, and numerous clinical trials are ongoing. However, most candidates have not achieved statistically significant improvements in patient survival. To address these limitations, nanotechnology-enabled cancer nanovaccine platforms have attracted increasing attention. Relative to conventional vaccine formats, cancer nanovaccines can improve antigen delivery efficiency and promote robust antitumor immune responses. Among these approaches, cell membrane nanoparticle (CNP)-based vaccines have been widely investigated and show strong biocompatibility and immunogenicity. Notably, preservation of major histocompatibility complex (MHC)-associated peptide repertoires on CNPs can help leverage endogenous antigen presentation and may reduce reliance on *in silico* tumor antigen

Abbreviations: 201M, gp100:209-217; AAM, Antibody-anchored membrane; AI, Artificial intelligence; AI-cell, Artificial Immune Cell; AiLNP, Artificial ICD LNP; APC, Antigen-presenting cell; ApoA1, Apolipoprotein A1; ATRA, All-trans-retinoic acid; BBB, Blood-brain barrier; bIL-15, Membrane bound IL-15/IL-15 receptor α complex; CaP, Calcium phosphate; CCM, Cancer cell membrane; CCR, C-C motif chemokine receptor; CD, Cluster of differentiation; CDX-110, Rindopepimut; cGAS, Cyclic GMP-AMP synthase; circRNA, Circular RNA; CLX, Celecoxib; CNN, Convolution Neural Network; CNP, Cell membrane nanoparticle; COMET, Composite Material Transformer; COX-2, Cyclooxygenase-2; CRT, Calreticulin; CTL, Cytotoxic T lymphocyte; CVB3, Coxsackievirus B3; CXCR4, C-X-C motif chemokine receptor 4; DAMP, Damage-associated molecular pattern; DC, Dendritic cell; dLN, Draining lymph node; DNN, Deep Neural Network; DOX, Doxorubicin; DP, Diphenylalanine; EBV, Epstein-Barr virus; ECM, Extracellular matrix; EGFRvIII, Epidermal growth factor receptor variant III; ER, Endoplasmic reticulum; ESC, Embryonic stem cell; ESO, Esomeprazole; FBS, Fetal bovine serum; GA, Gambogic acid; GBM, Glioblastoma; GI, Gastrointestinal; GM-CSF, Granulocyte-macrophage colony-stimulating factor; GNN, Graph Neural Network; gp100, Glycoprotein 100; GSH, Glutathione; HDL, High-density lipoprotein; HER2, Human epidermal growth factor receptor 2; HMGB1, High-mobility group box 1; HNSCC, Head and neck squamous cell carcinoma; ICB, Immune checkpoint blockade; ICD, Immunogenic cell death; IDO, Indoleamine 2,3-dioxygenase; IFN, Interferon; IL, Interleukin; iNKT, Invariant natural killer T cell; irAE, Immune-related adverse effect; IV, Intravenous; IVIS, *In vivo* imaging system; JAK, Janus kinase; KO, Knockout; LiON, Likelihood-based Optimization of Nanoparticles; LLM, Large Language Model; LN, Lymph node; LNP, Lipid nanoparticle; LSTM, Long Short-Term Memory; LSTV, Large-scale structural tumor variant; M, Microfold cell; mAg, Membrane-associated antigens; MDSC, Myeloid-derived suppressor cell; MHC, Major histocompatibility complex; MLP, Multi-Layer Perceptron; MMP, Metalloproteinase; Mn-LDH, Mn-based layered double hydroxide; MOF, Metal-organic framework; MPL, Polybia-mastoparan O; MPLA, Monophosphoryl lipid A; MSN, Mesoporous silica nanoparticle; NGS, Next-generation sequencing; NIR, Near-infrared; NK, Natural killer cell; NLRP3, NOD-like receptor pyrin domain-containing protein 3; Nov-S, Norovirus S domain; NPC, Nasopharyngeal carcinoma; OMV, Outer membrane vesicle; OVA, Ovalbumin; PAMP, Pathogen-associated molecular pattern; PBMC, Peripheral blood mononuclear cell; PBPK, Physiologically Based Pharmacokinetic; PD-L1, Programmed death ligand 1; PEI, Polyethyleneimine; PEPA, Poly(ethylpropylaminomethyl methacrylate); PHTA, Poly(ortho-hydroxy tertiary amine); PLGA, Poly(lactic-co-glycolic acid); PNP, Polymeric nanoparticle; Poly(I:C), Polyinosinic-polycytidylic acid; Prz, Prazosin; R, Arginine; R848, Resiquimod; RF, Random Forest; RNN, Recurrent Neural Network; ROS, Reactive oxygen species; SBA, Soybean agglutinin; scFv, Single-chain variable fragment; scRNA-seq, Single-cell RNA sequencing; SHAP, SHapley Additive exPlanations; siRNA, Small interfering RNA; SIRP α , Signal regulatory protein α ; SNA, Spherical nucleic acid; SpA, Staphylococcal protein A; SpC, SpyCatcher; SPION, Superparamagnetic iron oxide nanoparticle; SpT, SpyTag; SR-B1, Scavenger receptor class B type 1; SRSF1, Serine/arginine-rich splicing factor 1; STAT, Signal transducer and activator of transcription; STING, Stimulator of interferon genes; SVM, Support Vector Machines; TAM, Tumor-associated macrophage; TAP, Transporter associated with antigen processing; TCR, T cell receptor; TDLN, Tumor-draining lymph node; TIM-3, T cell immunoglobulin and mucin domain-containing 3; TLIME, Tumor lymph node immune microenvironment; TLR, Toll-like receptor; TME, Tumor microenvironment; TNF- α , Tumor necrosis factor α ; β -AR, β -adrenergic receptor; β Gluc, β -Glucan.

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prediction alone, a longstanding bottleneck in cancer vaccine development. In parallel, artificial intelligence (AI) is driving rapid advances across biomedical engineering and is applied to cancer nanovaccine research, including antigen identification, nanoparticle design optimization, and immune response modeling. This review comprehensively summarizes recent progress in anticancer nanovaccine research, with a particular focus on the development of CNP-based cancer nanovaccines. It also reviews the emerging AI-enabled strategies relevant to this field, including antigen prediction, formulation and design optimization, and computational modeling of immune responses. Finally, an outlook on future research directions and remaining challenges in this rapidly evolving field is provided.

1 | Introduction

Cancer is characterized by uncontrolled cellular proliferation and marked molecular heterogeneity, including the presence of tumor antigens [1–3]. As tumor antigens arise from tumor-associated alterations and are absent from normal tissues, they can be recognized as non-self antigens and elicit strong immune responses. Mutated proteins can generate immunogenic epitopes that are processed and presented to the immune system, enabling immune surveillance and, in some contexts, tumor cell elimination [4, 5]. Tumor antigens are typically captured by antigen-presenting cells (APCs), which process and present antigenic peptides. In particular, tumor antigen-derived peptides are presented to cluster of differentiation 8⁺ (CD8⁺) T cells via major histocompatibility complex class I (MHC-I) molecules, thereby initiating adaptive immune responses [6, 7]. However, many tumors not only evade immune recognition by downregulating MHC-I expression, but also engage immunosuppressive pathways—the upregulation of programmed death ligand 1 (PD-L1) and CD47—which signal through signal regulatory protein α (SIRP α) on myeloid cells [8–13]. Accordingly, the use of tumor antigens as cancer vaccines represents a central strategy in cancer immunotherapy [14].

Cancer vaccination aims to elicit coordinated immune responses that recognize and eliminate cancer cells expressing non-self antigens. Following vaccination, antigenic material is captured primarily by dendritic cells (DCs) [15]. Antigens are internalized through diverse DC surface receptors, including scavenger and lectin receptors, and are commonly administered with adjuvants. During DC maturation, the expression of co-stimulatory molecules (e.g., CD80 and CD86), MHC molecules, and the chemokine receptor C-C motif chemokine receptor 7 (CCR7) increases, a process that also promotes DC trafficking to draining lymph nodes (dLNs). As DCs migrate, they process internalized antigen and predominantly present peptides via MHC class II molecules to CD4⁺ T cells, while cross-presentation enables loading onto MHC-I molecules to prime CD8⁺ T cells [16–18] (Figure 1).

Within this framework, the development of tumor antigen-based cancer vaccines remains a major focus of cancer immunotherapy research. Progress in this area has been accelerated by next-generation sequencing (NGS) and bioinformatics, which enable systematic identification of candidate antigens [3]. These candidates can be broadly classified as shared antigens (observed across patients) or personalized, patient-specific antigens derived from unique somatic mutations [19, 20]. Nevertheless, many

tumor antigen-based vaccines have shown limited clinical efficacy. This is in part because they share key limitations of conventional peptide vaccines, including short in vivo half-life, rapid clearance, and insufficient immunogenicity. These constraints can limit APC activation, weaken T cell priming, and reduce therapeutic benefit [21, 22]. Nanotechnology has therefore emerged as a promising approach to improve vaccine performance.

Nanotechnology-based vaccines offer several advantages over conventional peptide formulations. Antigens can be encapsulated in or displayed on nanoparticles composed of polysaccharides, proteins, lipids, or other organic and inorganic materials [23]. These formulations can protect antigens from degradation, prolong systemic circulation, and reduce rapid clearance [24]. Among available platforms, lipid-based carriers have been extensively developed for protein and nucleic acid delivery. They support surface functionalization (e.g., with targeting ligands) and allow fine control of nanoparticle size and surface charge through lipid composition [25]. Such physicochemical tuning, combined with appropriate adjuvants, can enhance APC targeting and lymph node (LN) accumulation, thereby promoting robust immune responses [26, 27]. Co-delivery of antigen and adjuvant is particularly important for converting immunologically cold tumors into inflamed, T cell-infiltrated tumors and for improving overall immunotherapeutic efficacy [28, 29]. Despite these advances, clinical translation of nanomedicine remains limited, in part because protein corona formation can compromise in vivo stability, biodistribution, and efficacy.

In contrast to synthetic lipid nanoparticles (LNPs), cell membrane nanoparticles (CNPs) are intrinsically biocompatible and can reduce protein corona formation and premature clearance by the mononuclear phagocyte system [30]. Moreover, CNPs can inherit targeting properties from their source cells, enabling endogenous, cell type-derived interactions. CNPs generated from cancer cell membranes can therefore serve dual roles as tumor-targeted delivery vehicles and as antigen carriers for immune activation [31, 32]. Tumor antigens displayed on tumor-cell membrane MHC-I complexes can be recognized by APCs, potentially eliciting potent antitumor immune responses. This nanovaccine strategy may be further advanced through integration with ex vivo immune cell expansion methods and improved tumor antigen prediction pipelines.

In this review, we summarize the protein- and mRNA-based cancer vaccines currently under clinical evaluation to define their key advantages and limitations. We then discuss cancer

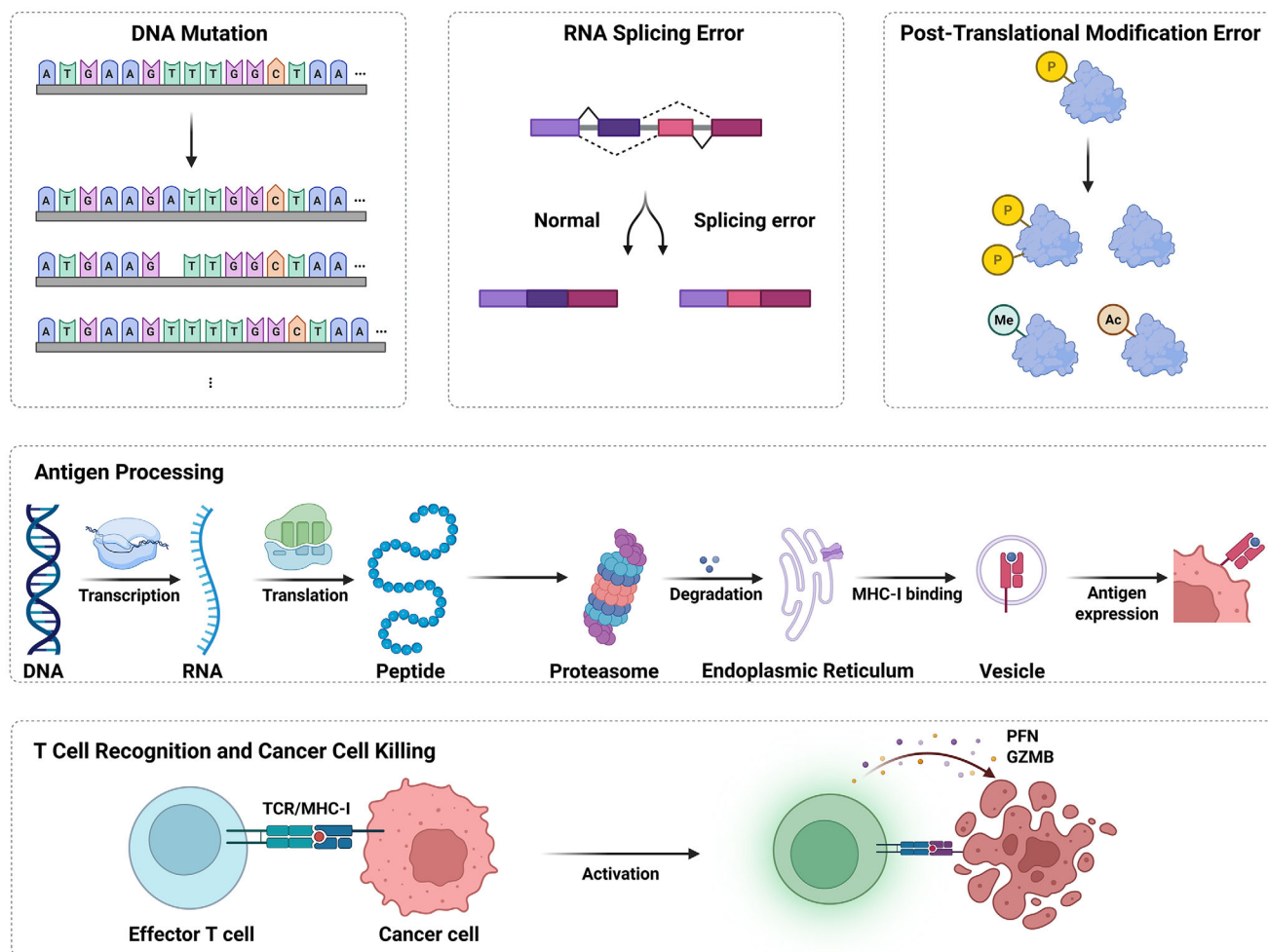


FIGURE 1 | Generation of tumor antigens, antigen expression, and T cell activation. Tumor antigens arise from altered proteins produced by tumor genomic and transcriptomic abnormalities, including single-nucleotide substitutions, insertions and deletions, aberrant RNA splicing, and dysregulated post-translational modifications. These proteins are degraded by the proteasome, and the resulting peptides are transported into the endoplasmic reticulum, loaded onto MHC-I molecules, and displayed on the tumor cell surface. T cells recognize peptide–MHC complexes, become activated, and induce tumor cell death. Created in BioRender. Noh, I. (2025) <https://BioRender.com/il7rqhz>.

nanovaccines as a strategy to retain the strengths of cancer vaccination while addressing the limitations of conventional formulations. To reflect recent developments, we highlight concepts and representative findings from cancer nanovaccine studies published in recent years, with a focus on CNP-based nanovaccines. Furthermore, we summarize tumor antigen prediction and delivery vehicle development using artificial intelligence (AI) technologies. Finally, we outline the remaining challenges and discuss future opportunities for applying AI in cancer vaccine and cancer nanovaccine development.

2 | Cancer Vaccines and Cancer Nanovaccines

2.1 | Tumor Antigen Vaccines, and AI-Assisted Antigen Prediction

Tumor antigen-based vaccines have advanced alongside high-throughput sequencing approaches, particularly NGS. These technologies enable the rapid profiling of tumor transcriptomes and exomes, supporting the identification of candidate tumor

antigens and immunogenic epitopes for vaccine development, including shared targets and, when relevant, patient-specific candidates [33, 34]. Integration with immunopeptidomics refines the selection of biologically relevant candidates, while single-cell RNA sequencing (scRNA-seq) addresses the challenges of intratumoral heterogeneity in uncovering cancer antigens. Moreover, the advent of AI-assisted epitope prediction has significantly enhanced the precision and diversity of vaccine design [35–38].

2.1.1 | Peptide-Based Cancer Vaccines

Peptide-based vaccines represent one of the earliest therapeutic cancer vaccine formats. Tumor antigens encompass self-derived antigens that are overexpressed, aberrantly expressed, or differentiation-associated in tumors; in some contexts, shared tumor-specific mutant epitopes have also been targeted. During malignant progression, tumor cells can accumulate genetic alterations and dysregulated RNA processing, which may generate aberrant protein products and antigenic peptides [39] (Figure 2).

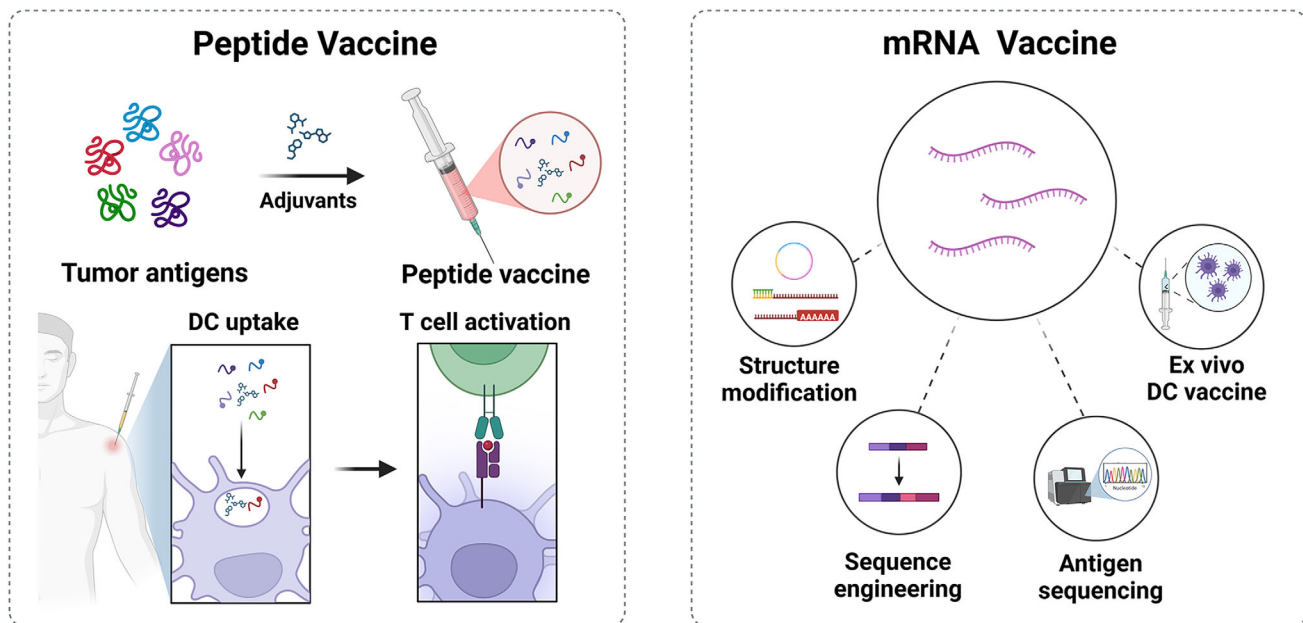


FIGURE 2 | Peptide- and mRNA-based cancer vaccines. Tumor antigen-based cancer vaccines combine antigen peptides with adjuvants to elicit antitumor immunity. After uptake by DCs, these formulations promote T cell priming and activation. mRNA cancer vaccines deliver mRNA encoding tumor antigens and are administered either by direct injection or via DC-based vaccination following ex vivo DC loading/activation. Chemical and structural optimization of mRNA can improve stability and translation, while NGS supports tumor antigen identification and sequence design; inclusion of multiple antigen sequences can further enhance immunogenicity. Created in BioRender. Noh, I. (2025) <https://BioRender.com/vnb7qw6>.

These peptides are produced by intracellular processing, transported into the endoplasmic reticulum (ER), loaded onto MHC-I molecules, and displayed at the cell surface for immune surveillance [40]. Although T cells are generally tolerant to self-antigens, tumor antigens can be recognized in a proinflammatory context, enabling T cell priming and effector responses [41]. This rationale underpins the development of tumor antigen-directed peptide vaccines [42].

Glycoprotein 100 (gp100), a melanocyte-associated protein, is overexpressed in melanoma. The gp100:209–217 (210 M) peptide is a modified epitope derived from amino acids 209–217 of gp100 that binds human leukocyte antigen HLA-A*02:01 and has been evaluated in multiple clinical trials. However, phase III studies did not demonstrate a significant survival benefit, highlighting the need for additional immunologic support to improve the clinical efficacy of tumor antigen-based peptide vaccines [43–45].

Glioblastoma (GBM) remains among the most difficult malignancies to treat, in part because the blood-brain barrier (BBB) restricts therapeutic access, making immunotherapy-based strategies attractive. However, the immunosuppressive tumor microenvironment (TME), together with the BBB, limits both drug delivery and immune cell infiltration [46].

Rindopepimut (CDX-110) targets epidermal growth factor receptor variant III (EGFRvIII), a tumor-specific mutant protein expressed in a subset of GBMs. This vaccine comprises an EGFRvIII-derived peptide conjugated to keyhole limpet hemocyanin and showed encouraging results in phase II trials but did not improve overall survival in a subsequent phase III trial [47, 48]. Antigen escape, including reduced EGFRvIII expression after

treatment, was also observed [49]. These findings support the need for combination strategies, including immune checkpoint blockade (ICB), and the incorporation of multiple antigens to mitigate immune evasion.

For ovarian and breast cancers, therapeutic vaccines targeting human epidermal growth factor receptor 2 (HER2) are under development. HER2 is overexpressed in these malignancies and is a validated target of antibody-based therapies. Nelinepimut-S is a peptide vaccine based on the immunogenic E75 epitope of HER2 and is commonly administered with immune adjuvants such as granulocyte-macrophage colony-stimulating factor (GM-CSF) [50]. Although vaccination can enhance antigen-specific immune responses, phase III trials did not show a significant improvement in overall survival, consistent with findings from other tumor antigen-directed peptide vaccine programs [51].

In renal cell carcinoma, a multi-peptide vaccine strategy (IMA901) has been evaluated. IMA901 comprises 10 peptides, including 9 CD8⁺ T cell epitopes and 1 CD4⁺ T cell epitope. When cyclophosphamide was administered before vaccination to reduce regulatory T cell activity, enhanced antitumor immune responses were reported. However, despite these immunologic effects, phase III trials did not demonstrate improved overall survival [52].

Collectively, these studies underscore the ability of cancer vaccines to induce antitumor immune responses, while also highlighting persistent limitations, including suboptimal antigen delivery and insufficient magnitude or durability of immunity. These challenges have motivated extensive development of cancer nanovaccines that use nanotechnology-based deliv-

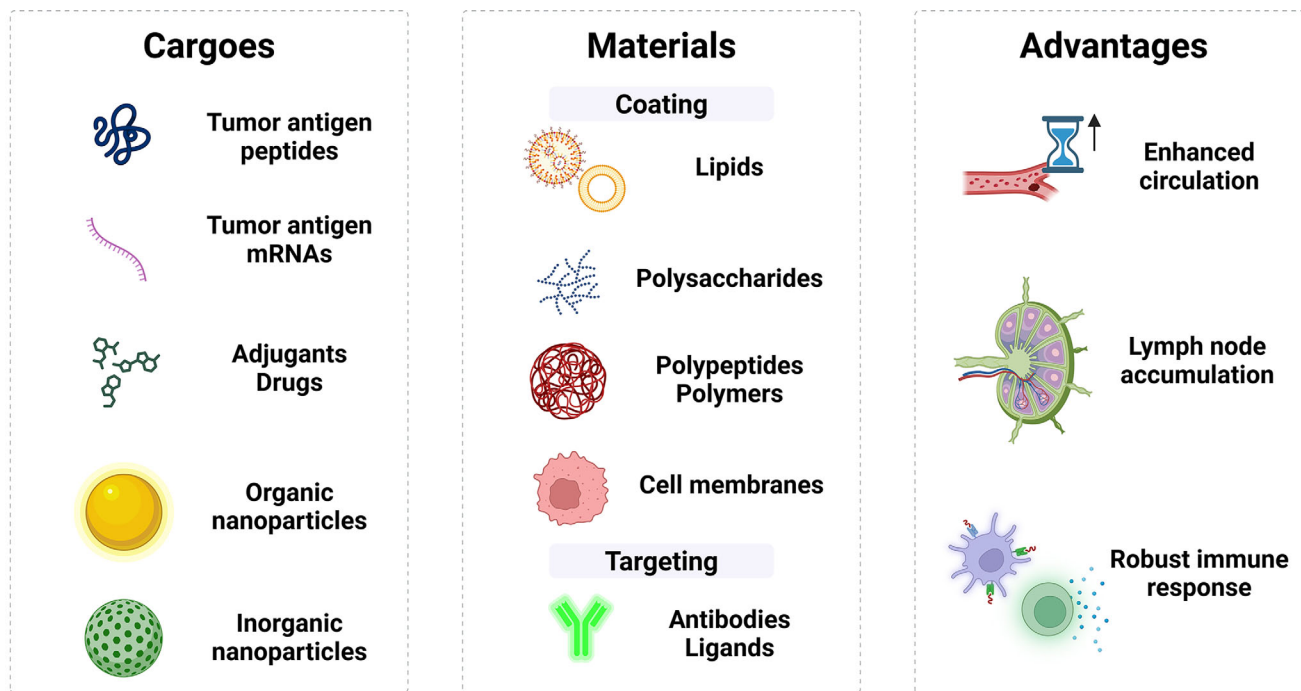


FIGURE 3 | Nanoparticle-based cancer nanovaccines. Cancer nanovaccines use nanotechnology to address limitations of conventional vaccine platforms by improving antigen protection, delivery, and immune engagement. In addition to established cargoes (antigen peptides, mRNA, and adjuvants), nanovaccines can incorporate materials that confer structural stability and tunable function. These systems can be engineered for surface coating and targeting to prolong circulation, enhance LN accumulation, and elicit potent antitumor immune responses. Created in BioRender. Noh, I. (2026) <https://BioRender.com/7qz6dkj>.

ery systems to improve vaccine pharmacology and immune priming.

2.1.2 | MRNA-Based Cancer Vaccines

mRNA-based cancer vaccines can address a key limitation of peptide vaccines, namely, low immunogenicity arising from inefficient cross-presentation [53, 54]. These vaccines elicit antitumor immunity by delivering synthetic mRNA that encodes shared tumor antigens or patient-specific antigens. mRNA platforms are attractive because they enable rapid sequence design and scalable manufacturing, and they can encode multiple antigens to enhance immunogenicity [55] (Figure 2).

After uptake by APCs, particularly DCs, mRNA is translated in the cytoplasm. The resulting proteins are processed by the proteasome and presented on MHC-I, promoting CD8⁺ T cell priming and activation [56]. In many settings, this mechanism yields stronger cytotoxic T lymphocyte (CTL) responses than tumor antigen-based peptide vaccines, which often rely on MHC-II presentation of exogenous antigens and preferentially expand CD4⁺ T cell responses [57]. In addition, mRNA can engage innate immune sensors, including Toll-like receptors (TLRs), thereby inducing inflammatory cytokines that further support adaptive immunity [58, 59].

mRNA vaccines can be administered via direct injection or ex vivo DC vaccination [60, 61]. Direct delivery routes include intradermal, subcutaneous, intravenous, intramuscular, and intratumoral

administration. Following injection, mRNA is taken up by DCs and other local cells, and antigen-bearing APCs traffic to dLNs, where they initiate T cell priming [62, 63].

However, naked mRNA is rapidly degraded by extracellular RNases and shows poor cellular uptake owing to limited membrane permeability [64]. Accordingly, cationic polymers and nanoparticle-based carriers are commonly used to improve stability and delivery. In ex vivo approaches, patient-derived DCs are transfected with mRNA (often by electroporation) to promote DC activation prior to reinfusion [65].

mRNA vaccines encoding shared antigens can be manufactured in advance, enabling rapid treatment initiation and reducing the time and cost relative to fully personalized products. For example, BNT111 is a fixed-combination mRNA vaccine encoding four melanoma-associated antigens (NY-ESO-1, MAGE-A3, tyrosinase, and TPTE). In phase I studies, BNT111 induced T cell responses as monotherapy and in combination with ICB [66].

Advances in NGS have enabled individualized mRNA vaccines based on patient-specific somatic mutations [67]. Tumor sequencing supports the identification of candidate antigens, followed by the design of mRNA constructs encoding the selected neoepitopes. KEYNOTE-942 is a phase IIb study evaluating mRNA-4157 in combination with pembrolizumab in resected melanoma. The combination therapy showed improved 18-month recurrence-free survival compared to pembrolizumab alone, with reductions in metastasis and death, and increased antigen-specific CD8⁺ and

TABLE 1 | Summary of cancer nanovaccines.

Delivery carrier	Cancer type	Cancer cell line	Antigen type	Active ingredients	AI-assist	Refs.
Lipid (Liposome)	- Melanoma	- B16F10	- gp100 peptide	- CLX	- No	- [85]
	- Renal carcinoma	- B16-OVA	- Tumor antigen mRNA	- R848, CpG-1826	- No	- [90]
	- Lymphoma	- B16-EBNA3A	- OVA peptide	- QS-18	- No	- [86, 88]
	- Colorectal adenocarcinoma	- Renca	- Trunc-LMP2A peptide	- Man9GlcNAc2-Asn, CpG	- No	- [92]
	- Lung carcinoma	- E.G7-OVA	- Nes2LR peptide	- PD-L1 siRNA	- No	- [87]
	- HNSCC	- CT26, MC38	- OVA peptide	- ATRA	- No	- [89, 133]
		- MC38-OVA	- Tumor antigen mRNA	- circRNA	- No	- [90]
		- CT26-HER2	- Cancer cell mRNA	- MPLA	- No	- [94]
		- LLC	- OVA mRNA	- CpG	- No	- [95]
		- UM-SCC-104	- HER2 mRNA		- No	- [96]
			- MAGE-A1		- No	- [93]
			- E7 peptide		- No	- [98]
	- Melanoma	- B16-OVA	- Membrane peptide	- Ionizable lipid	- No	- [99]
	- Colorectal adenocarcinoma	- B16F10	- OVA mRNA	- α -galactosylceramide, IL-15 mRNA, ESO, STAT3 siRNA	- No	- [101, 104, 106, 107]
	- Lung epithelial cell	- CT26	- OVA peptide		- Yes	- [134]
PLGA	- Lung carcinoma	- TC-1	- TRP2 mRNA	- Ionizable lipid	- No	- [102]
	- Cervical adenocarcinoma	- LLC	- Membrane Peptide	- Mannose	- No	- [103]
		- HeLa	- E7 peptide	- Mannose	- No	- [99]
			- Membrane peptide	- Ionizable lipid	- No	- [100, 135]
			- mRNA	- LLOMe, Mn ²⁺	- No	- [105]
				- CRT, HMGB1	- Yes	- [136]
				- Ionizable lipid		
	- Melanoma	- B16BL6-OVA	- OVA peptide	- Riboxim	- No	- [110]
	- HCC	- B16-OVA	- OVA peptide	- CpG	- No	- [112]
		- Hepal-6-hGPC3	- GPC3 peptide	- HMGB1	- No	- [137]
	- Melanoma	- B16-OVA	- OVA peptide	- PEI-4BImi-Man, ODEX	- No	- [114]
	- Lymphoma	- E.G7-OVA	- OVA mRNA	- β -glucan	- No	- [113]

(Continues)

TABLE 1 | (Continued)

Delivery carrier	Cancer type	Cancer cell line	Antigen type	Active ingredients	AI-assist	Refs.
Polypeptide Synthetic Polymer	- Lymphoma	- E.G7-OVA	- OVA peptide	- 3RDP(F5)	- No	- [121]
	- Melanoma	- B16-OVA	- OVA mRNA	- PHTA	- No	- [115]
	- Colorectal adenocarcinoma	- CT26-LMP2	- OVA peptide	- PEPA, MSA-2	- No	- [120]
			- LMP2 mRNA	- PEG ^[TK PF]	- No	- [116]
Metal	- Melanoma	- B16F10	- Cytosolic DNA	- CaCO ₃	- No	- [118]
	- Colorectal adenocarcinoma	- B16-OVA	- OVA peptide	- Zn-MOF, CpG, Mn ²⁺	- No	- [117, 119]
		- CT26	- Cytosolic DNA	- Synthetic adjuvant	- Yes	- [138]
	- Breast cancer	- 4T1-PD1	- HER2/neu peptide	- CaCO ₃	- No	- [118]
				- Synthetic adjuvant	- Yes	- [138]

CD4⁺ T cell responses [68].

Compared to protein-based antigens, mRNA vaccines can achieve higher immunogenicity and straightforward multiplexing. Ongoing studies have focused on improving mRNA stability and expression. Circular mRNA is more resistant to degradation than linear mRNA, which can prolong the antigen production. Structural optimization of the 5' and 3' ends, including capping and poly(A) tail engineering, can further increase expression and reduce mRNA degradation. Additional strategies aim to enhance DC priming by co-encoding immunomodulators, such as cytokines and T cell costimulatory ligands, alongside antigens [69–73].

Despite these advantages, several limitations remain to be addressed. Predicted antigens may not be efficiently processed and presented on MHC molecules after expression, or they may not be recognized by cognate T cell receptors (TCRs). Personalized vaccine development also imposes time and cost constraints on the process [74]. Most critically, the intrinsic instability of mRNA results in rapid degradation in vivo and suboptimal delivery efficiency. Consequently, integration with nanotechnology-based delivery systems is essential to fully realize the potential of mRNA-based cancer vaccines.

2.1.3 | AI-Assisted Antigen Prediction

Conventionally, the selection of antigens for cancer vaccine development is a time- and labor-intensive process, relying on fundamental bioinformatics and direct experimental validation. Although significant funds have been invested to identify optimal cancer antigens, sufficient clinical efficacy has not been demonstrated to date. AI-driven antigen screening and design have emerged to address these needs.

Initially, AI-driven antigen prediction models, such as Multi-Layer Perceptron (MLP), were utilized to identify or design epitopes that bind to the MHC. These models predict binding affinity by processing digitized amino acid sequences. However, they are constrained by fixed input dimensions and a limited capacity to evaluate immunogenicity within specific sequence contexts [75]. Currently, MLP architectures are primarily integrated as substructures for specialized computational tasks within more complex frameworks.

Convolutional Neural Network (CNN)-based models were subsequently introduced to identify local motifs within peptide sequences. Compared to MLP architectures, CNNs demonstrate superior performance in extracting patterns from learned antigen data, enabling the systematic identification of significant MHC-binding sequence motifs [76]. For instance, FIONA2, developed for antigen discovery from large-scale structural tumor variants (LSTVs), integrates a 1D CNN model with NetMHCpan and NetMHCIIpan MLP architectures. This platform successfully captured MHC-I and MHC-II binding epitopes to predict antigen sequences for MC38 tumor cells. Nevertheless, such models require large-scale RNA sequencing for each sample and face

limitations in capturing long-range correlations in the extended sequences [77, 78].

To address sequence length variability, Recurrent Neural Network (RNN) and Long Short-Term Memory (LSTM)-based prediction models were developed. These architectures maintain sequential context and extend predictive capabilities beyond simple binding affinity to estimate the kinetic stability—specifically the half-life— of pMHC complexes. Notably, they facilitate the identification of TCR binding potential and the likelihood of immune response activation [79].

Transformer architectures and attention-based models, including Large Language Models (LLMs), have emerged to facilitate both the discovery and de novo design of MHC-binding epitopes. By processing amino acid sequences similarly to natural language processing, transformer-based platforms excel at deciphering relatively long epitopes and key immunological motifs, which are challenging for conventional CNN or RNN frameworks. HLApollo, a Transformer-based model, predicts pMHC-I presentation. By integrating protein language models, it demonstrates high pan-allelic generalization without separate gene expression data, offering improved clinical relevance for rare alleles [80]. Similarly, ImmuneApp-Neo utilizes attention mechanisms to analyze large-scale immunopeptidomics and enhances the accuracy of ligandome analysis across multiple alleles through sophisticated deconvolution. Notably, it overcomes data scarcity by using pretrained models and offers high interpretability, moving away from the “black-box” nature of earlier models [81].

Graph Neural Network (GNN) models are specifically optimized to predict the interactions between epitopes, pMHCs, and TCRs through 3D structural modeling based on features extracted from Transformer models. GraphMHC precisely simulates the stereostructural fitness between the receptor pocket and epitope by generating graphs from amino acid sequences. By simulating the attraction and repulsion of residues to autonomously identify structural features, this model achieves superior predictive accuracy compared with conventional frameworks [82].

In conclusion, AI has become an indispensable tool in cancer vaccine development, with a diverse range of antigen prediction models emerging through the synergistic combination of multiple architectures. This evolution, from simple affinity calculations to highly accurate frameworks, reflects the improved ability of AI to identify the optimal sequences required for effective vaccination. Although challenges such as data bias toward specific genotypes and high computational requirements remain, the continuous advancement of AI architectures and the accumulation of large-scale clinical databases will drive the design of next-generation personalized cancer vaccines.

2.2 | Nanoparticle-Based Cancer Nanovaccines and AI-Assisted Formulation Optimization

Nanoparticle-based cancer nanovaccines often exhibit improved antitumor efficacy compared to free tumor

antigen vaccines or physically mixed components, as they enable the coordinated co-delivery of antigens and immunomodulators [23, 83]. By enhancing antigen presentation and promoting localized and concentrated adjuvant delivery, nanoparticles can increase the magnitude and quality of vaccine-elicited antitumor immune responses [84] (Figure 3 and Table 1).

2.2.1 | Liposome and LNP-based Cancer Nanovaccines

Liposomes and LNPs are established delivery systems for proteins and nucleic acids, including mRNA. These carriers can incorporate hydrophobic drugs and immunomodulators within lipid compartments while delivering tumor antigens or antigen-encoding nucleic acids. Antigen sequences can be delivered as complexes with cationic lipids or by encapsulating nucleic acids within the lipid phase. These strategies address key limitations of nucleic acid vaccines, including poor in vivo stability and inefficient cellular uptake, thereby improving intracellular delivery and antigen expression.

2.2.1.1 | Liposome. A liposome-based cancer nanovaccine incorporating the B16F10 melanoma antigen gp100 and the cyclooxygenase-2 (COX-2) inhibitor celecoxib (CLX) inhibited tumor growth in B16F10 models when administered as a combination regimen. In a combination strategy comprising a DC vaccine, lipid-conjugated gp100, and CLX-loaded liposomes, treatment increased TIL abundance and interferon-gamma (IFN- γ) secretion, resulting in tumor growth suppression and improved overall survival [85].

Resiquimod (R848), a TLR7/8 agonist, potently activates CD8⁺ T cells but can cause systemic toxicity when administered in conventional formulations. In B16-OVA models, ProLNG-001 liposome combined with ovalbumin (OVA) peptide antigen elicited robust activation of T cells and natural killer (NK) cells and suppressed tumor growth [86]. Liposomes have also been used to improve the therapeutic index of saponin adjuvants. QS-18, a saponin related to QS-21 (used in AS01), has strong adjuvant activity but is limited by hemolytic toxicity and other adverse effects [87].

Collectively, liposome platforms can enhance the accumulation of vaccine cargos and immunomodulators, and they enable incorporation of potent agents that would otherwise be constrained by toxicity, thereby improving the antitumor efficacy of cancer vaccines.

Leveraging the in vivo stability of liposomes, active delivery can be achieved by encapsulating magnetic materials to enable field-guided targeting. In one approach, B16-OVA cells were treated with HOCl to increase immunogenicity, and tumor lysates plus CpG-1826 were co-encapsulated with γ -Fe₂O₃ superparamagnetic iron oxide nanoparticles (SPIONs) in liposomes to generate the Ag-MLs vaccine [88]. To enhance cross-presentation and MHC-I antigen presentation from exogenous antigens, a pH-responsive liposome (SBA-pH-lip) was developed incorporating the polysaccharide MGlu-Dex and a soybean agglutinin (SBA)-derived glycopeptide (Man₉GlcNAc₂-Asn) to target DC lectin receptors [89].

Liposome platforms have also been applied to mRNA cancer vaccines. In CT26, MC38, and B16F10 models, lipopolyplex nanovaccines consisting of tumor antigen mRNA complexed with protamine and loaded into liposomes inhibited tumor growth. Therapeutic efficacy depended on administration parameters, with stronger effects reported after multi-site subcutaneous injection at higher concentrations. These formulations elicited robust tumor-directed CD8⁺ T cell responses, showed activity in therapeutic, prophylactic, and anti-metastasis settings, and exhibited synergy with anti-PD-1 antibodies. No toxicity was reported within the effective dose range. In a phase I clinical trial, two patients receiving the lipopolyplex treatment showed tumor growth suppression and induction of antitumor immune responses [90].

Epstein-Barr virus (EBV) is linked to multiple malignancies, motivating efforts to develop EBV antigen-based immunotherapies [91]. In nasopharyngeal carcinoma (NPC) models, liposomal vaccines loaded with full-length LMP2 mRNA induced tumor-specific immune responses [92].

The mLPR nanovaccine co-encapsulates the TLR4 agonist monophosphoryl lipid A (MPLA) within the hydrophobic membrane and MAGE-A1 antigen mRNA in the aqueous core. This formulation activated APCs through TLR4/janus kinase 2 (JAK2)/signal transducer and activator of transcription 1 (STAT1) signaling. In preclinical studies, mLPR suppressed lung tumor growth and reduced bone metastasis, promoted cytokine secretion (including IFN- γ and IL-12), and induced NK cell-mediated antibody-dependent cellular cytotoxicity [93].

A cationic liposome vaccine co-encapsulating PD-L1 small interfering RNA (siRNA) and total RNA from MC-38 tumor cells reduced PD-L1 expression in DCs and MC-38 tumors while enhancing CD8⁺ T cell cytotoxicity. These cationic liposomes inhibited autophagy, thereby improving MHC-I-mediated antigen presentation by APCs, and accumulated in tumors and tumor-draining lymph nodes (TDLNs) [94] (Figure 4F).

Liposomes incorporating all-trans-retinoic acid (ATRA) showed efficient uptake by DCs and enhanced mRNA expression. In an MC38-OVA orthotopic model, intramuscular administration of ATRA-containing liposomes activated CD8⁺ T cells and induced expression of the gut-homing receptors (CCR9 and integrin $\alpha 4\beta 7$, thereby imprinting gut tropism [95] (Figure 4A).

Circular mRNA (circRNA) has greater intrinsic stability than linear mRNA, enabling prolonged protein expression. A nanovaccine (panCAR-VAC) was generated by loading chimeric antigen receptor (CAR) circRNA into LNPs. In vivo depletion of NK cells or macrophages reduced antitumor efficacy, indicating that both populations contributed to therapeutic activity [96].

The temperature-sensitive nanoimmunomodulator NIL-IM-LIP was developed to remodel the tumor lymph node immune microenvironment (TLIME). NIL-IM-LIP delivers pH-responsive NGR motifs and matrix metalloproteinase-2 (MMP2)-responsive IL-15 from tumors to LNs. Subsequent near-infrared (NIR) irradiation induces immunogenic cell death (ICD) via IR-780 and delivers the indoleamine 2,3-dioxygenase (IDO) inhibitor 1-MP, while concurrently suppressing regulatory T cells. This strategy

activated intratumoral CD8⁺ T cells and NK cells, produced antitumor effects in both immunologically “hot” B16F10 and “cold” CT26 models, and showed synergy with anti-PD-1 therapy [97] (Figure 4E).

Studies of spherical nucleic acid (SNA) nanovaccines further underscore the importance of structural design. In SNA formulations comprising liposomes encapsulating CpG DNA and the E7 peptide antigen, inclusion of a strong anchor (CpG-(C12)₉) improved LN accumulation and enhanced CD8⁺ T cell activation relative to a weaker CpG-cholesterol anchor [98].

2.2.1.2 | LNPs. For personalized cancer vaccine development, LNPs have been engineered to encapsulate membrane-associated antigens (mAg) derived from resected tumors, including B16-OVA and CT26. Following construction of an ionizable lipid library, formulations were optimized for particle size and DC activation. These LNPs improved antitumor efficacy in both B16-OVA and CT26 models, and co-administration with anti-PD-1 further enhanced therapeutic responses, supporting the feasibility of generating personalized nanovaccines from resected tumors and the potential utility of ICB in colorectal cancer [99].

LNPs co-delivering Mn²⁺ (a cofactor of stimulator of interferon genes (STING)) and an E7 peptide activated the cyclic GMP-AMP synthase (cGAS)-STING axis in DCs, enhanced TIL function, and elicited cervical cancer-specific immune responses. Combination with ICB further improved antitumor efficacy, consistent with a synergistic strategy that couples STING activation with ICD-oriented therapy [100]. To improve targeting, mRNA LNPs incorporating α -galactosylceramide, a ligand for invariant natural killer T (iNKT) cells, increased both antibody responses and CD8⁺ T cell activation [101].

To target CD206 on APCs, “sweet” LNPs (STLNPs) were functionalized with monosaccharides (glucose, galactose, or mannose) and loaded with OVA-encoding mRNA. Among these variants, mannose-decorated particles (STLNP-Man) showed the strongest DC targeting, with an LNP formulation containing 15% mannose-lipid achieving favorable particle size and mRNA delivery efficiency [102].

In a related mannose-lipid strategy, a hydrazone-linked mannosylated LNP (Hyd-Man LNP) was developed to enhance endosomal escape. Under acidic endosomal conditions, hydrazone cleavage generates amines that undergo protonation and promote endosomal disruption. Hyd-Man LNP improved delivery efficiency relative to controls, with particularly increased mRNA expression. In the B16F10 model, antigen-encoding mRNA delivery improved survival and enhanced immune responses [103]. (Figure 4B–D).

LNPs formulated with tertiary amine headgroups (derived from piperazine) and unsaturated linolenic acid-derived tails preferentially accumulated in the spleen, which is advantageous for reprogramming splenic DCs. In this context, LNPs co-encapsulating antigen mRNA and mRNA encoding a membrane-bound IL-15/IL-15 receptor α complex (bIL-15) delivered both antigen and IL-15 signals to support T cell responses. This LNP showed therapeutic and prophylactic activity across multiple tumor models [104].

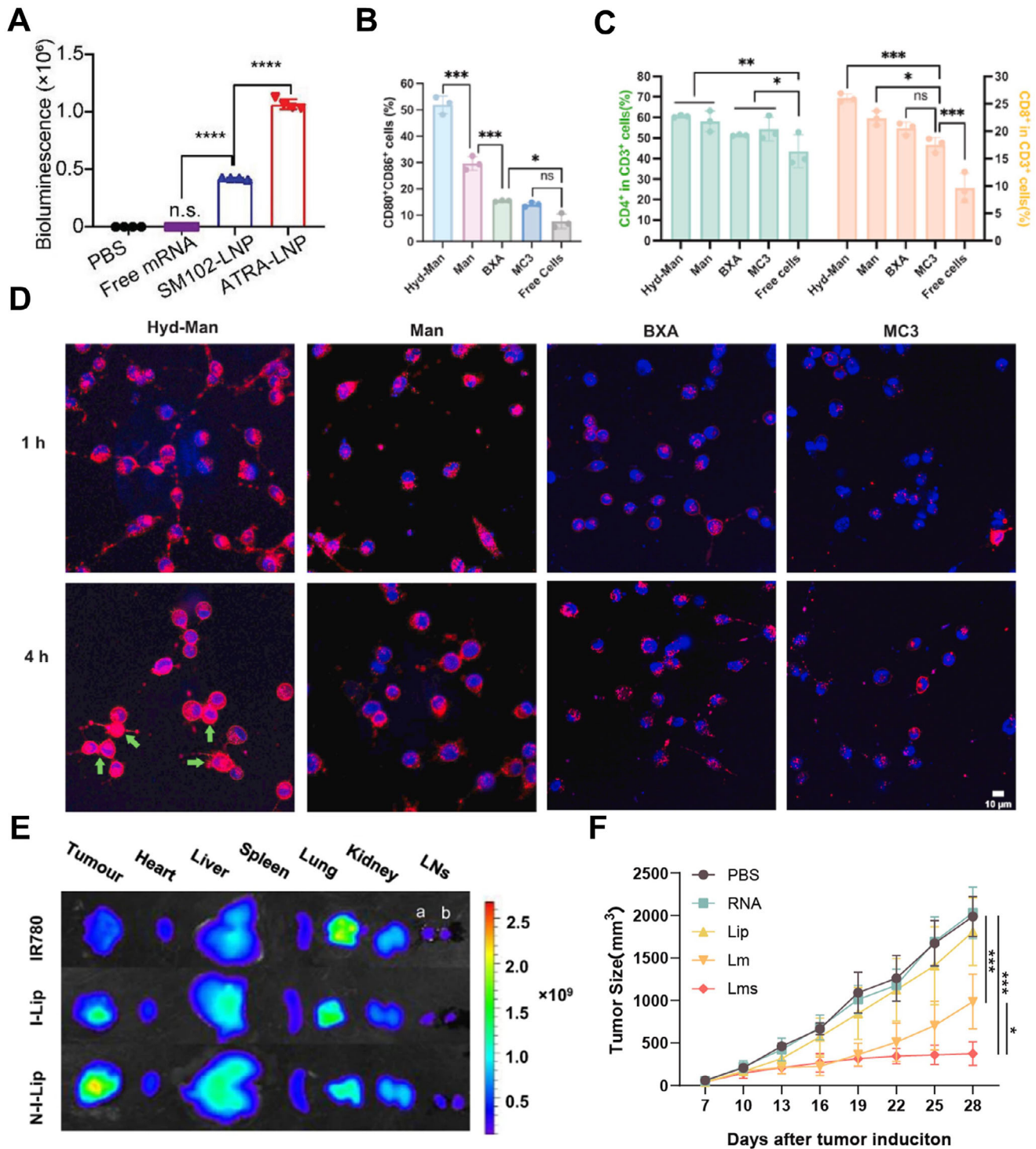


FIGURE 4 | Preclinical therapeutic efficacy evaluation of cancer nanovaccines. (A) Transfection efficiency of free mRNA, SM102-LNP, and ATRA-LNP ($N/p = 15$) in DC2.4 cells ($n = 4$). Reproduced under the terms of the CC-BY Creative Commons Attribution 4.0 International license (<https://creativecommons.org/licenses/by/4.0>) [95]. Copyright 2024, Wiley-VCH. DC activation in vitro. (B) Flow cytometry analysis of DC cells after treated with each TRP₂₁₈₀₋₁₈₈ mRNA-encapsulated LNPs for 24 h ($n = 3$). (C) Flow cytometry analysis of T cells after co-incubated with DCs treated with indicated TRP₂₁₈₀₋₁₈₈ mRNA-encapsulated LNPs for 24 h ($n = 3$). Significant differences were assessed using a one-way ANOVA with Tukey test. Data are presented as mean $\pm$ SD. (D) Cy5-labeled mCherry mRNA encapsulated different LNPs (red) uptake was measured at 1 h and 4 h. Scale bar = 10 μ m. Reproduced under the terms of the CC-BY Creative Commons Attribution 4.0 International license (<https://creativecommons.org/licenses/by/4.0>) [103]. Copyright 2025, Elsevier. (E) Ex vivo imaging of the tumor, organs and LNs at 48 h. I-Lip: IR780-Lip; N-I-Lip: NGR-IR780-Lip. Reproduced under the terms of the CC-BY Creative Commons Attribution 4.0 International license (<https://creativecommons.org/licenses/by/4.0>) [97]. Copyright 2023, Springer Nature. (F) MC38 tumor growth inhibition study. Data are presented as the mean $\pm$ SD (ns (not significant), * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$). Reproduced under the terms of the CC-BY Creative Commons Attribution 4.0 International license (<https://creativecommons.org/licenses/by/4.0>) [94]. Copyright 2025, Elsevier.

TABLE 2 | Comparison between nanoparticle-based cancer nanovaccines and CNP-based cancer nanovaccines.

	Nanoparticle-based cancer nanovaccines	CNP-based cancer nanovaccines
Antigen source	Synthetic peptides mRNAs	Surface pMHC repertoire
Composition	Lipids Polymers Inorganic materials	Cancer cell membranes
Delivery mechanism	Endocytosis	Endocytosis Membrane fusion
Immune activation	Requires intracellular antigen processing	Direct antigen presentation Preservation of native antigen structures
Targeting ability	Relatively low Usually requires ligand modification	Relatively high Intrinsic targeting ability
Biocompatibility	Material-dependent May requires surface modification	Relatively high Prolonged circulation time

Artificial ICD LNPs (AiLNPs) decorated with ICD-associated signals, including calreticulin (CRT) and high mobility group box 1 (HMGB1), effectively activated APCs. When loaded with cancer cell membrane proteins, AiLNPs targeted and activated splenic DCs and elicited stronger antitumor activity and immune activation (including CD8⁺ T cells) than formulations containing ICD markers alone or membrane proteins alone [105]. Esomeprazole (ESO)-loaded mRNA LNPs have also been reported to enhance endosomal and lysosomal escape, improving cytosolic delivery and increasing mRNA expression relative to ESO-free LNPs. These formulations further increased extracellular matrix (ECM) penetration and enhanced intratumoral immune responses [106].

The heterocyclic lipid component activated STING signaling and increased type I interferon production, resulting in enhanced tumor growth inhibition, reduced regulatory T cells and myeloid-derived suppressor cells (MDSCs), and increased CD8⁺ T cell responses [107].

Overall, liposomal and LNP platforms enable stable delivery of diverse antigen formats (including proteins and mRNA) and can enhance cellular uptake, antigen expression, and cross-presentation to promote robust CD8⁺ T cell-mediated antitumor immunity. Co-encapsulation of immunomodulators can further improve efficacy by reprogramming the tumor immune microenvironment, activating APCs, and enabling synergy with ICB.

However, formulation-dependent variables, including lipid structure, targeting ligands, and stimulus-responsive chemistries, can increase manufacturing complexity and compromise batch-to-batch reproducibility, often requiring substantial optimization effort and cost. In addition, intrinsic liabilities of liposomes and LNPs, such as non-specific biodistribution and preferential accumulation in the liver and spleen, motivate continued development of targeting strategies, careful toxicity management, and rigorous evaluation of long-term safety.

2.2.2 | Polymeric and Inorganic Nanoparticle-Based Cancer Nanovaccines

Poly(lactic-co-glycolic acid) (PLGA) is a U.S. FDA-approved polymer widely used in drug delivery, including vaccine formulations. PLGA is biocompatible and biodegradable and has favorable mechanical properties and manufacturing flexibility, supporting broad pharmaceutical utility [108]. PLGA nanoparticles can encapsulate diverse antigens and immunomodulators, including peptides and nucleic acids, and can promote sustained immune activation through controlled-release kinetics [109].

A PLGA nanovaccine co-encapsulating the OVA peptide and Riboxim enhanced activation of murine and human DCs and increased tumor-specific CD8⁺ T cell responses. This formulation inhibited primary tumor growth and metastasis and improved overall survival [110].

PLGA nanovaccines functionalized with polyethyleneimine (PEI) and loaded with the TLR3 agonist polyinosinic-polycytidylic acid (poly(I:C)), together with patient-derived antigens, increased expression of DC activation markers (CD80, CD83, and CD86) and elevated T cell activation (CD69) and IFN- γ expression in peripheral blood mononuclear cells (PBMCs) [111]. PLGA functionalized with hr-8, a peptide targeting the DC surface receptor dendritic and thymic epithelial cell-205 (DEC-205), and loaded with PLGA@Ag/CpG enhanced DC activation and CD8⁺ T cell responses against B16-OVA melanoma compared with OVA/CpG or PLGA@OVA/CpG [112].

Overall, PLGA nanoparticles can elicit potent and durable antitumor immunity by enabling co-delivery of antigens and adjuvants with sustained release and by supporting modular functionalization for combination regimens. However, the lack of inherent targeting underscores the need for surface engineering, including targeting ligands or functional coatings, to maximize delivery efficiency.

TABLE 3 | CNP-based cancer nanovaccines.

Membrane functionalization	Cancer type	Cancer cell line	Active ingredients	Refs.
- Normal	- Melanoma - Colon adenocarcinoma - Breast cancer	- B16F10 - CT26 - MC38 - 4T1	- CpG, mRNA, cGAMP, SN21-DSPE-PEG, 22A peptide - GA - Prz, <i>E. coli</i> OMV - CpG	- [151, 153, 155, 156] - [149, 161] - [150, 165] - [152, 161]
- Drug Treatment	- Melanoma - Breast cancer - Lung epithelial cell	- DOX-treated B16F10 - SRSF1 and Pladienolide B-treated 4T1 - DOX-treated 4T1 - CVB3-treated 4T1 - MTO and curcumin-treated TC-1	- Upregulated MHC-I, R848 - Splicing error mRNA, Ce6 - Upregulated MHC-I, R848 - Upregulated CRT, Downregulated PD-L1, B7-H3, and CD47 - DAMPs	- [157] - [154] - [157] - [159] - [158]
- Hybridization	- Colon adenocarcinoma - Breast cancer - HNSCC	- MC38, <i>E. coli</i> OMV - 4T1, <i>E. coli</i> OMV, Ginseng-derived EV - SCCVII, DC2.4	- LPS - LPS, Ginseng-derived EV - MPLA	- [166] - [167, 168] - [162]
- Chemical Engineering	- Melanoma - Lymphoma	- Enzymatically desialyated B16-OVA - Anti-CD28 conjugated E.G7-OVA	- TIM siRNA - MPLA	- [169] - [170]
- Genetic Engineering	- Melanoma - Colon adenocarcinoma - PDAC	- CD47KO/CRT-expressing B16F10 - Anti-CD40 scFV-expressing MC38 - Anti-CD40 scFV-expressing KPC, Panc02	- CpG - Tumor lysate - Tumor lysate	- [171] - [172] - [172]

*)To date, no AI-based studies have utilized CNP for cancer nanovaccines.

β -Glucan (β Gluc) and mRNA-loaded LNPs (β Gluc/mRNA@LNP) enable oral delivery of vaccine cargos to immune cell-rich regions of the gastrointestinal (GI) tract. After oral administration, β Gluc/mRNA@LNP efficiently accumulates in DCs and microfold (M) cells, bypasses hepatic first-pass uptake, and delivers cargo to mesenteric LNs to promote T cell activation. In in vivo study, this oral nanovaccine showed superior antitumor efficacy relative to intravenous (IV) administration and offers potential advantages in patient convenience and adherence [113].

Sustained antigen exposure can improve vaccine efficacy by supporting prolonged immune priming. Nanoparticles in 4-arm PEG-ONH₂ and oxidized dextran hydrogel were released over time, taken up by DCs, and transported to LNs. In the B16-OVA model, an 8% gel formulation produced greater tumor growth inhibition than comparator conditions, consistent with a benefit from sustained antigen release [114].

Polymeric nanoparticle (PNP) vaccines based on poly(ortho-hydroxy tertiary amine) (PHTA) were developed to provide controlled innate stimulation with effective mRNA delivery. Among tested variants, PHTA-C18 showed favorable mRNA packaging, delivery efficiency, and prolonged plasma half-life. Relative to LNPs, PNPs exhibited enhanced LN accumulation and increased expression of DC activation markers (CD80, CD86, and MHC-I), supporting robust antitumor immune responses [115].

Fluorinated PEI reduced serum-mediated clearance and showed therapeutic efficacy in models of nasopharyngeal carcinoma, consistent with self-adjuvant activity. Fluorinated PEI maintained approximately 50% transfection efficiency under 50% fetal bovine serum (FBS) conditions and promoted DC activation via TLR4 [116].

Metal-organic frameworks (MOFs) are tunable nanomaterials with high surface area and porous architectures that enable co-loading of antigens and adjuvants. ZPM@OVA-CpG, a mannose-coated zinc MOF loaded with OVA and CpG, targeted DCs in LNs and degraded under acidic conditions in lysosomes and within the TME [117].

Resected CT26 or B16F10 tumors were heat-inactivated, lysates were prepared, and biomineralized using CaCl₂ and Na₂CO₃ to generate TLC@CaCO₃. This autologous cancer nanovaccine activated the cGAS-STING pathway, consistent with contributions from tumor antigens and cytosolic DNA [118].

An Mn-based layered double hydroxide (Mn-LDH) loaded with OVA self-adjuvant system promoted DC activation through intracellular glutathione (GSH) depletion and STING pathway activation. Mn-LDH degradation consumed GSH and released Mn²⁺ and antigen. During degradation, reactive oxygen species (ROS) were generated, and ROS, together with Mn²⁺-mediated STING activation, enhanced DC activation and inhibited B16-OVA tumor growth [119].

The PEIM-OVA nanovaccine, comprising acid-ionizable poly(ethylpropylaminomethyl methacrylate) (PEPA), iron oxide nanoparticles, the STING agonist MSA-2, and OVA,

enhanced STING activation in APCs within dLNs. Relative to soluble antigen, PEIM-OVA increased CD8⁺ T cell activation by 55-fold, reduced lung metastasis, and induced tumor regression [120].

Immunostimulatory peptides composed of arginine (R) and fluorinated diphenylalanine (DP) can self-assemble with antigens into nanoparticulate structures. Self-assembly and antigen affinity are tunable through peptide length and degree of fluorination. Among the tested designs, 4RDP(F5) showed the highest binding affinity for OVA, enhanced DC activation, and strong CD8⁺ T cell responses were induced in an OVA-expressing E.G7 lymphoma model [121].

Collectively, diverse bio-, inorganic-, and polymer-based delivery systems, including β -glucan, protein- or peptide-derived nanostructures, hydrogels, MOFs, and metallic or polymeric nanoparticles, enable flexible control over antigen stability, targeting, and presentation. Leveraging carrier-intrinsic properties can further tailor innate activation and adaptive priming to promote antitumor immunity.

However, the expanding diversity of delivery platforms introduces challenges in scalable and reproducible manufacturing, control of physicochemical properties, and stability of antigen and adjuvant loading, which together increase the complexity of development. In addition, depending on the material composition (e.g., metal ions, ROS-generating components, or cationic polymers), these systems may introduce toxicity, excessive immune stimulation, or tissue-specific adverse effects, underscoring the need for rigorous and comprehensive safety evaluations.

2.2.3 | AI-Assisted Nanoparticle Design

Recently, the rising necessity to increase the efficiency of nanoparticle development has led to the exploration of AI-assisted nanoparticle design has been explored to predict complex interactions between multiple constituents. To address the toxicity and instability inherent in conventional cationic lipid carriers with complicated lipid components in LNPs, a hybrid model integrating XGBoost, Random Forest (RF), and Support Vector Machines (SVM) was developed based on large-scale HTS data [122]. Furthermore, SHapley Additive exPlanations (SHAP) have been used to precisely calculate the contribution of each lipid component by back-calculating the optimal ratios for specific cell types [123]. Notably, this approach facilitates the expansion of interpretable AI models by addressing the “black-box” limitation of traditional deep learning. CNN models have also been used to predict in vitro nanoparticle release profiles [124].

Generative AI has become prominent in optimizing formulations and designing individual components. Likelihood-based Optimization of Nanoparticles (LiON), a generative deep learning model, was designed to develop novel synthetic lipids for mRNA delivery. LiON overcame human bias and saved time and labor compared to traditional HTS systems [125]. Composite Material Transformer (COMET) model focuses on formulation optimization. Based on datasets of LNP formulations and cell

lines, COMET calculates optimal formulations using parameters such as components, relative molar ratios, N/P ratios, and mixing conditions. Remarkably, it predicts both delivery efficiency and post-freeze stability, and shows application potential for toxicity assessment [126].

Beyond material development and formulation optimization, various AI models have been developed to assess the in vivo properties of nanoparticles. Unlike in vitro environments, nanoparticles encounter complex biological interactions that reduce delivery efficiency and may induce adverse effects. Hybrid models combining Deep Neural Networks (DNN) and Physiologically Based Pharmacokinetic (PBPK) frameworks have been developed to predict drug release profiles and pharmacokinetics. These models showed improved accuracy in short-term (within 24 h) tumor delivery efficiency and other organ accumulation. This model enables researchers to predict the metabolism and biodistribution of nanoparticles without relying entirely on in vivo imaging systems (IVIS) or fluorescence analysis of tissue sections entirely [127–130].

Transformer and Graph Neural Network (GNN) architectures have been developed to elucidate the interactions between nanoparticles and the immune system. The Artificial Immune Cell (AI-cell) model predicts the immune activation capability of nucleic acid nanoparticles. This model drastically reduced the time required to measure in vitro immune cell activation and cytokine levels from several days to 2–4 s [131, 132].

In conclusion, similar to antigen design, AI technologies have been applied to predict nanoparticle fabrication, biodistribution, toxicity, and biological interactions. These AI-driven approaches have significantly enhanced the efficiency of nanocarrier development for cancer nanovaccines, and are emerging as potential strategies to increase the success of next-generation cancer therapeutics.

3 | CNP Cancer Nanovaccines

Unlike conventional polymer-based nanoparticles, CNPs are derived from biological cell membranes and offer several advantages over synthetic delivery systems. The plasma membrane is an organic bilayer composed of phospholipids, membrane proteins (including diverse receptors), and polysaccharides. It mediates cellular metabolism and signal transduction while retaining the key phenotypic features of the source cell [139].

CNPs were introduced by the Zhang group through erythrocyte membrane-coated drug delivery systems. These biomimetic nanoparticles can achieve preferential delivery to source cells or tissues through homotypic targeting, which is sometimes referred to as the biointerference effect. This property allows CNP to deliver their cargo via direct membrane fusion, not only through endocytosis, as in conventional nanoparticles [140]. In addition, CNPs exhibit prolonged in vivo circulation and reduced susceptibility to performance losses associated with protein corona formation, immune clearance, and nonspecific immune activation. The low zeta potential and cell membrane proteins (e.g., integrins, cadherins, and CDs) on the CNP surface confer

unique stability under physiological conditions and enhanced targeting capability. Compared to conventional nanoparticles, CNPs increase targeted drug delivery efficiency. Most administered nanoparticles accumulate in liver and kidneys, but CNPs show lower organ accumulation. Notably, in a patient-derived xenograft model study, CNPs exhibited higher targeting efficiency toward autologous patient-derived cells than those from other patients with the same cancer [30, 141–143] (Figure 5). These unique properties of CNPs offer several advantages for developing cancer nanovaccines.

On the surface of cancer cells, antigens are already natively expressed as pMHC complexes. Cancer cell CNPs directly activate T cells or are integrated into the cell membranes of APCs through trogocytosis. This mechanism, known as ‘cross-dressing’, enables the induction of potent immune responses without the requirement for conventional cross-presentation [144, 145].

Peptide and mRNA vaccines require intracellular processing for MHC loading, whereas CNPs directly present antigens to APCs, preserving the 3D structural integrity of the pMHC complex. Furthermore, the diverse antigen repertoire expressed on the CNP surface prevents antigen escape triggered by tumor heterogeneity, which is a major cause of failure in cancer vaccine therapy [146, 147]. This cross-dressing mechanism can also overcome the low immunological activity epitope problems, despite the predicted binding affinity.

Furthermore, CNPs possess the inherent advantage of delivering various co-stimulatory molecules with antigens. Specifically, the diverse proteins and polysaccharides on the cell membrane modulate immune signaling pathways, the complexity of which cannot be fully replicated by current AI frameworks. This underscores the potential of CNPs as a superior and powerful alternative to conventional synthetic nanoparticle platforms (Table 2). Extensive work has leveraged cancer cell-derived membranes for anticancer drug delivery and cancer nanovaccine development [32, 148] (Table 3).

3.1 | Natural Cancer Cell Membrane-Based Cancer Nanovaccines

In a study using the CT26 colorectal cancer cell line, PLGA nanoparticles loaded with gambogic acid (GA) and coated with cancer cell membranes (CCM-PLGA/GA NPs) enhanced DC activation, increased tumor accumulation, and improved tumor growth inhibition relative to uncoated PLGA/GA NPs [149].

A liposomal hybrid nanovaccine (prazosin (Prz)@LINV) co-delivering MC38 tumor cell membrane and Prz induced ICD, as indicated by increased CRT exposure and release of HMGB1 and ATP. Prz@LINV improved intracellular delivery in MC38 and DC2.4 cells relative to Prz@LIP and increased cytotoxicity toward MC38 cells. Mechanistically, Prz inhibited autophagy, thereby augmenting ER stress-associated ICD and enhancing MHC-I-mediated antigen presentation [150].

A nanovaccine comprising mesoporous silica nanoparticles (MSNs) loaded with the β -adrenergic receptor (β -AR) inhibitor propranolol and CpG, followed by coating with B16F10 melanoma

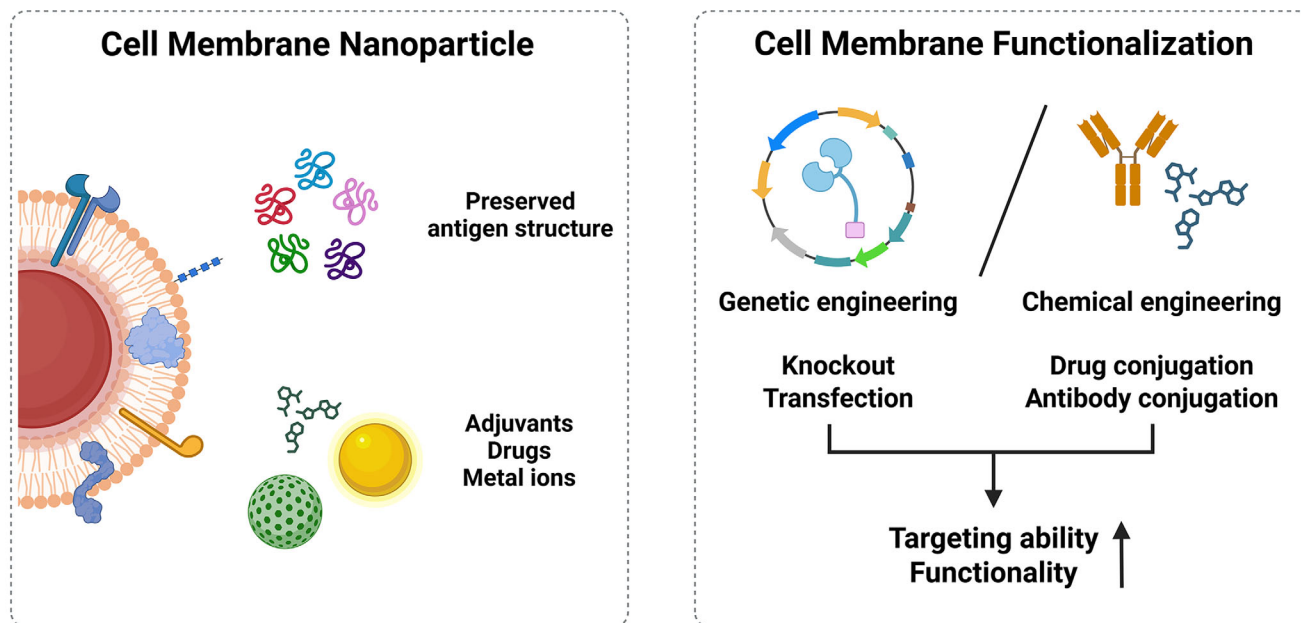


FIGURE 5 | Potential of CNP technology as a cancer nanovaccine platform. Unlike conventional cancer vaccines and nanovaccines, CNPs can preserve tumor-derived antigens displayed on MHC-I molecules at the cancer cell surface. This structural preservation supports presentation of a diverse antigen repertoire and can promote robust antitumor immune responses. CNPs also retain key advantages of nanovaccine platforms by enabling incorporation of diverse functional materials. In addition, the membrane surface can be further engineered through genetic or chemical modification, providing enhanced and multifunctional performance relative to conventional cancer nanovaccines. Created in BioRender. Noh, I. (2025) <https://BioRender.com/cjaiq4p>.

cell membranes, showed enhanced LN accumulation and activation of APCs. Propranolol release inhibited β -AR signaling, promoting migration and activation of naïve and effector T cells within LNs while reducing immunosuppressive populations, including regulatory T cells, M2 macrophages, and MDSCs [151] (Figure 6E,F).

In addition to conventional $\alpha\beta$ T cells, $\gamma\delta$ T cells are enriched in epithelial tissues and participate in both innate and adaptive immunity. Their rapid activation and potent effector functions make them attractive targets for epithelial vaccine delivery, including microneedle-based approaches. In one study, mice received microneedles loaded with CpG-containing liposomes coated with 4T1 cancer membranes. Relative to subcutaneous injection, microneedle delivery increased $\gamma\delta$ T cell abundance in the epidermis and dermis and markedly elevated the fraction of activated DCs migrating to dLNs. This outcome was attributed to more efficient delivery of antigens and adjuvants to epithelial immune cells, whereas subcutaneous injection primarily targets adipose tissue with fewer resident immune cells and abundant fibroblasts and adipocytes. The microneedle vaccine increased tumor infiltration by CD27⁺, perforin⁺, and Ki67⁺ $\gamma\delta$ T cells and reduced regulatory T cells and MDSCs [152].

To broaden tumor antigen coverage, mRNA cancer vaccines can deliver multiple antigen-encoding mRNA sequences simultaneously. Core nanoparticles assembled from mRNA, the cationic peptide RALA, and PLGA were subsequently coated with membranes derived from activated DCs (ADM-mRNA-NPs). These nanoparticles accumulated efficiently in LNs and showed improved uptake by LN-resident DCs relative to conventional

PLGA nanoparticles, consistent with homotypic targeting. ADM-mRNA-NPs also supported prolonged antigen expression compared with PLGA-LNPs. In B16F10-OVA tumor-bearing mice, this platform improved tumor growth control and strengthened anti-tumor immune activation relative to PLGA-based formulations [153].

Cancer cells frequently exhibit aberrant mRNA splicing, generating diverse antigen repertoires. A nanovaccine formulated using the 4T1 cell membrane and mRNA showed potent prophylactic antitumor efficacy. To test whether perturbing splicing could further improve performance, splicing was disrupted in 4T1 cells using a Serine/arginine-rich splicing factor 1 (SRSF1) virus and the splicing modulator pladienolide B, and nanovaccines were then prepared from these cells. Vaccination with splicing-perturbed membranes improved tumor growth suppression and increased TILs, supporting the concept that deliberate splicing disruption can potentiate antitumor immunity [154].

A nanovaccine (MSN2@cGAMP@CM-SN21) was generated by coating MSNs loaded with cGAMP with B16F10 membranes and incorporating the macropinocytosis-inducing peptide SN21 conjugated to DSPE-PEG. This formulation enhanced delivery of membrane-associated antigens and cGAMP through C-X-C motif chemokine receptor 4 (CXCR4)-mediated endocytosis facilitated by SN21. The combination of B16F10-derived antigens and STING activation by cGAMP induced complete therapeutic responses in B16F10 tumors [155].

High-density lipoprotein (HDL) is actively transported into lymphatics via scavenger receptor class B type I (SR-BI) on

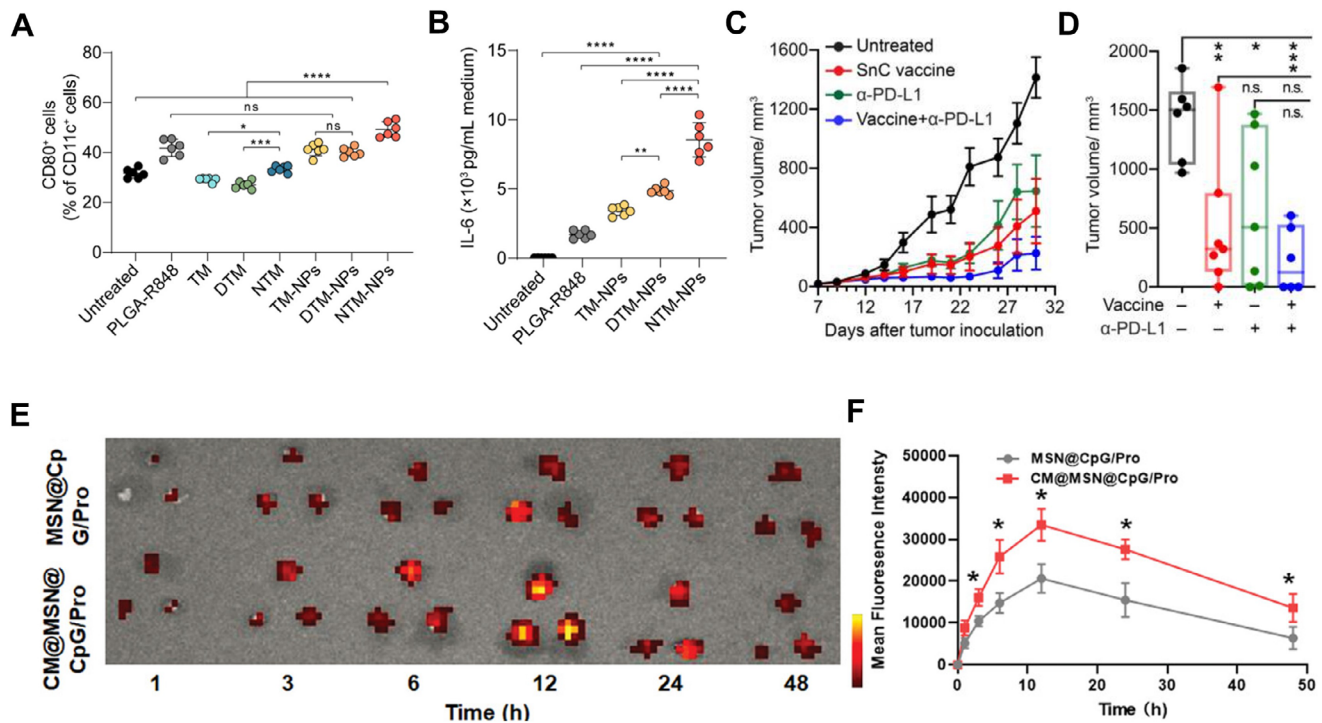


FIGURE 6 | Preclinical therapeutic efficacy evaluation of CNP-based cancer nanovaccines. NTM-NPs activated BMDCs' antigen uptake and maturation. (A) CD80 of BMDCs were confirmed by flow cytometry ($n = 6$). (B) IL-6 concentrations in the media were measured by ELISA ($n = 6$). Data are presented as mean $\pm$ SD. Statistical significance was analyzed using one-way ANOVA with Tukey's post hoc test. Reproduced with permission [157]. Copyright 2025, Cell press. (C) SnC vaccines induced antitumor immune responses and suppressed CT26 tumor growth. Data are presented as mean $\pm$ SEM. (D) CT26 tumor size at day 30. Paired t -test. For statistical analysis, *ns* (not significant), $*p < 0.05$, $**p < 0.01$, $***p < 0.001$. Reproduced with permission [161]. Copyright 2023, BMJ Publishing Group Ltd. (E) Fluorescence imaging of LNs after Cy5.5-labeled MSN@CpG/Pro and CM@MSN@CpG/Pro vaccination. (F) Quantitative analysis of fluorescence from Cy5.5-labeled MSN@CpG/Pro and CM@MSN@CpG/Pro in the LNs. Data are presented as mean $\pm$ SD, $n = 3$, $*p < 0.05$ versus the MSN@CpG/Pro group). Reproduced with permission [151]. Copyright 2023, Wiley-VCH.

LN endothelial cells. A nanovaccine (R837/LNPM-ML) was developed by loading the TLR7/8 agonist R837 into LNPs coated with cancer cell membranes, followed by incorporation of an apolipoprotein A1 (ApoA1)-mimetic 22A peptide. Although the 22A peptide induced unfavorable changes in particle size and charge, it increased LN accumulation and DC uptake. In the acidic endosomal environment, R837 was released, activated TLR7/8 signaling, promoted antigen presentation, and increased intratumoral CD8⁺ T cell infiltration and proliferation [156].

To enhance CNP nanovaccine efficacy, PLGA nanoparticles loaded with R848 were coated with tumor membranes derived from 4T1 and B16F10 tumors after treatment with doxorubicin (DOX) liposomes. DOX liposome-treated tumor cells showed increased expression of multiple ICD markers relative to untreated or free-DOX-treated cells. MHC-I expression was significantly upregulated, whereas PD-L1 levels were unchanged. CNPs generated from DOX liposome-treated membranes (NTM-NPs) promoted BMDC maturation, as reflected by increased CD80 and CD86 and elevated IL-1 β , IL-6, and tumor necrosis factor α (TNF- α), together with enhanced antigen presentation. In tumor-bearing mice, NTM-NPs increased CD8⁺ T cell abundance in the spleen and TDLNs and increased secretion of IFN- γ and granzyme B. Tumor recurrence and metastasis were reduced, supporting the potential of personalized CNP nanovaccines

for postsurgical prevention of relapse and dissemination [157] (Figure 6A,B).

PLGA nanoparticles coated with chitosan were decorated with E7 peptides and combined with membranes from ICD-induced TC-1 cells generated by mitoxantrone and curcumin treatment. These TC-1 CNPs were incorporated into an alginate hydrogel containing ATP. ATP release recruited DCs, and activated DCs migrated to dLNs and promoted CD8⁺ T cell priming [158].

4T1 cells infected with Coxsackievirus B3 (CVB3) showed reduced surface expression of PD-L1, B7-H3, and CD47, together with increased exposure of CRT. Vaccination with CNPs activated RAW264.7 macrophages, tumor-infiltrating macrophages, and CD8⁺ T cells, supporting the potential of combining oncolytic virus-based strategies with cancer nanovaccines [159].

Cancer nanovaccines can also be generated from senescent cancer cells derived from autologous hematologic or solid malignancies. During senescence, tumor cells can express cytokines such as IFN- γ and TNF- α , enabling development of personalized vaccines without exogenous adjuvants. These nanovaccines showed synergistic antitumor activity when combined with ICB [160]. Senescent cancer cell nanovaccines generated using CT26 and 4T1 cells also induced T cell responses through DC activation [161] (Figure 6C,D).

A hybrid nanovaccine (Hy-M-Exo) combining head and neck squamous cell carcinoma (HNSCC)-derived exosomes and DC membrane vesicles exhibited enhanced LN targeting via CCR7, which supports DC homing to LNs. Tumor antigens and endogenous danger signals present in HNSCC-derived exosomes further activated APCs and elicited robust CD8⁺ T cell responses [162].

Gram-negative bacterial outer membrane vesicles (OMVs) contain pathogen-associated molecular patterns (PAMPs) [163, 164]. OMVs isolated from *Escherichia coli* deliver lipopolysaccharide (LPS) to the cytoplasm, inducing IL-1 β secretion through the NOD-like receptor pyrin domain-containing protein 3 (NLRP3) inflammasome pathway. OMVs activate APCs and can induce epigenetic reprogramming in hematopoietic progenitors, enhancing responsiveness to subsequent stimuli. This trained immunity phenotype was associated with expansion of the multipotent progenitor 3 (MPP3; Flt3⁻ CD48⁺ CD150⁻) subset with myeloid-biased granulocyte and macrophage potential. OMV priming 1 week before cancer vaccination enhanced antitumor immune responses [165].

Hybrid nanovaccines composed of bacterial membranes, tumor cell membranes, and liposomes incorporate endogenous immunostimulants such as LPS, which can improve antitumor efficacy. In the MC38 colorectal cancer model, these hybrids accumulated efficiently in DCs and macrophages within TDLNs and induced potent antitumor immunity [166]. Related hybrid systems based on 4T1 tumor membranes and *E. coli* membranes showed therapeutic activity against lung-metastatic 4T1 tumors [167]. Similarly, hybrids composed of ginseng-derived extracellular vesicle-like particles and cancer cell membranes enhanced antitumor responses through TLR4 pathway activation by ginseng-derived vesicles [168].

Collectively, the direct use of naturally derived membrane-, protein-, and glycan-based carriers, including cancer cell membranes, DC membranes, senescent cell membranes, bacterial OMVs, and exosomes, can integrate antigenic diversity, targeting capacity, and immunomodulatory function within a single platform. Biological membrane properties such as homotypic targeting, LN homing, and provision of PAMPs and damage-associated molecular patterns (DAMPs) can enhance immunogenicity, prolong circulation, and improve APC selectivity relative to synthetic nanoparticles, thereby supporting customizable and personalized vaccine development.

3.2 | Functionalized Cancer Cell Membrane-Based Cancer Nanovaccines

Direct engineering of cancer cell membranes can endow nanovaccines with intrinsic targeting functions and enhanced immunogenicity.

3.2.1 | Chemical or Enzymatic Engineering

Sialic acids promote tumor growth and metastasis and contribute to immune evasion through interactions with Siglec receptors on immune cells. To better leverage the antigenic repertoire of cancer cell membranes, a nanovaccine was prepared using desialylated

B16-OVA cell membranes combined with a liposomal adjuvant carrying T cell immunoglobulin and mucin domain-containing 3 (TIM-3) siRNA. This desialylated membrane nanovaccine elicited robust antitumor immunity and inhibited tumor growth and metastasis more effectively than the corresponding sialylated membrane formulation [169].

E.G7-OVA tumor cell membranes were pretreated with IFN- γ for 48 h to upregulate MHC-I expression and were then hybridized with MPLA-loaded liposomes. The resulting hybrid nanoparticles were further functionalized with anti-CD28 antibodies via thiol-maleimide coupling to generate CCM-MPLA-aCD28 nanovaccines. Relative to CCM-aCD28, which provides direct T cell costimulation, or CCM-MPLA, which primarily promotes indirect T cell activation via antigen-presenting cells, CCM-MPLA-aCD28 induced greater T cell proliferation and activation. Collectively, these results support the concept that nanovaccines designed to engage both DCs and T cells can achieve higher therapeutic efficacy than platforms that predominantly target a single immune compartment [170].

3.2.2 | Genetic Engineering

CD47 on tumor cells interacts with SIRP α on phagocytes to deliver a “don’t eat me” signal and thereby promotes immune evasion. Removing immune-inhibitory ligands can enhance the activity of tumor cell membrane-derived nanovaccines. A nanovaccine comprising PEI25K/CpG nanoparticles coated with CD47-knockout (CD47KO) and CRT-expressing B16F10 cell membranes promoted stronger activation of BMDCs and other immune populations than control formulations. Combination therapy with anti-PD-L1 further augmented the antitumor immune responses elicited by this CNP nanovaccine [171].

An antibody-anchored membrane (AAM) nanovaccine platform (Nano-AAM/CD40) was generated from Panc02, KPC, and MC38 tumor cells engineered to express an anti-CD40 single-chain variable fragment (scFv). Nano-AAM/CD40 inhibited tumor growth in both immunologically “hot” and “cold” tumor models, presented CD40 costimulatory signals on the cancer cell membrane, and exhibited LN targeting. Encapsulation of tumor lysate further increased antitumor efficacy [172] (Figure 7A–C).

A modular nanovaccine strategy based on the SpyCatcher (SpC)–SpyTag (SpT) system was also developed. HEK-293 cells expressing SpC were used as a membrane source to generate modular CNPs that can be redirected toward different cancers by exchanging SpT-tagged antigens [148, 173].

Related approaches have leveraged bacterial OMVs as modular scaffolds. In one design, bacteria were engineered to express Staphylococcal protein A (SpA) variants that bind anti-PD-L1 antibodies together with SpC to capture SpT-tagged tumor antigens. OMV nanovaccines constructed by conjugating embryonic stem cell (ESC)-derived SpT-tagged tumor antigens and anti-PD-L1 antibodies showed prolonged in vivo retention of both antigen and antibody. In MB49 bladder cancer and LLC

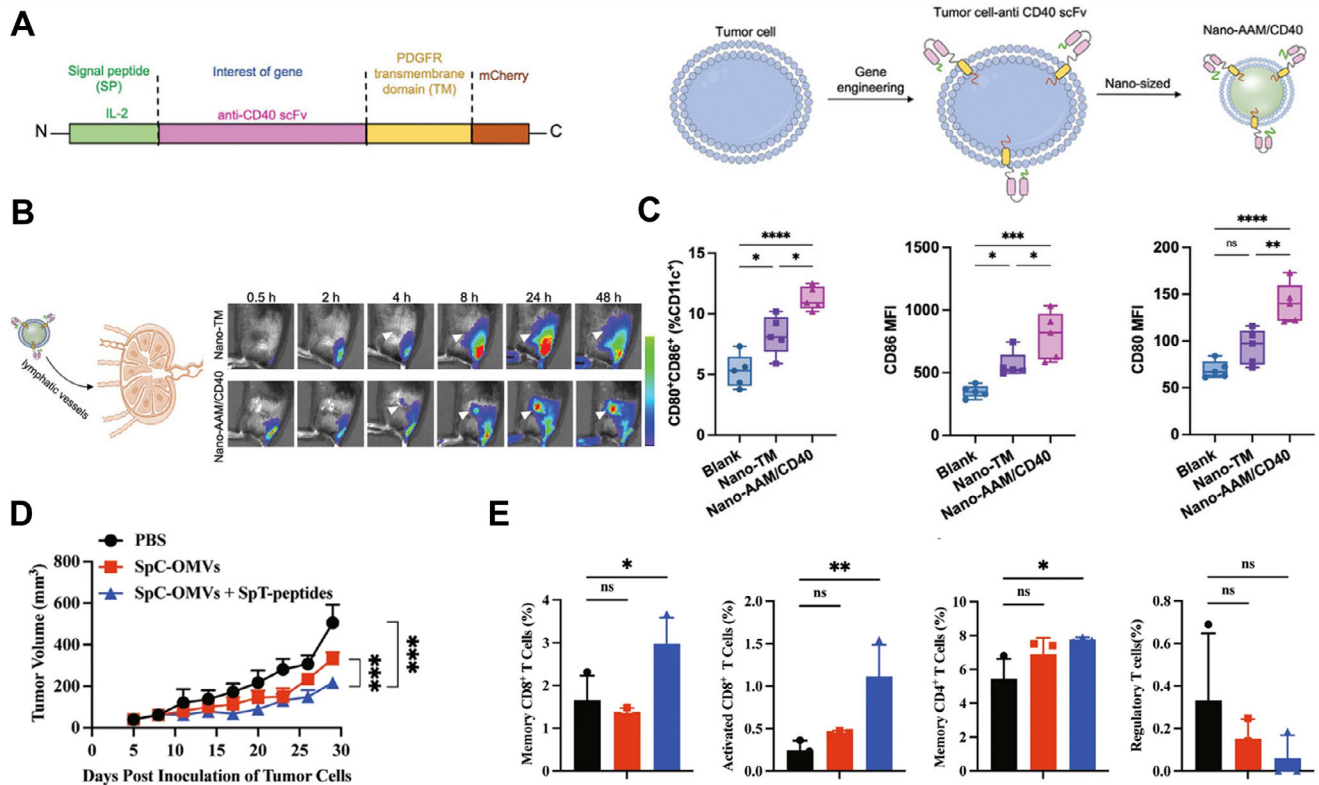


FIGURE 7 | CNP engineering for enhanced anti-tumor immune responses. (A) Genetic engineering of Nano-AAM/CD40. (B) In vivo fluorescence imaging of LN at the different time points after DiD-loaded Nano-TM or Nano-AAM/CD40 subcutaneous injection ($n = 3$). (C) Flow cytometry analysis of the in vivo DC maturation in the LNs after Nano-TM or Nano-AAM/CD40 treatment ($n = 5$). Data are presented as mean $\pm$ SD. Statistical analysis was performed using one-way ANOVA with Tukey's multiple comparisons test. Reproduced with permission [172]. Copyright 2023, Wiley-VCH. Tumor antigen-bound OMVs targeted tumors and inhibited bladder cancer growth. (D) Cancer growth inhibition of OMV-P. (E) Quantitative analysis of CD3⁺CD45⁺ T cells in LNs by flow cytometry ($n = 3$). Memory CD4⁺ T cells, CD4⁺CD44⁺ T cells. Memory CD8⁺ T cells, CD8⁺CD44⁺ T cells. Activated CD8⁺ T cells, CD8⁺Granzyme-B⁺ T cells. Treg, CD4⁺CD25⁺FoxP3⁺ T cells. Data are shown as mean $\pm$ SD. One-way ANOVA with a Tukey multiple comparisons test. ns (not significant), * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. Reproduced under the terms of the CC-BY Creative Commons Attribution 4.0 International license (<https://creativecommons.org/licenses/by/4.0>) [174]., Copyright 2024, Springer Nature.

lung cancer models that share antigens with ESCs, these OMV vaccines enhanced tumor targeting, inhibited tumor growth, and increased TILs [174] (Figure 7D–E). In parallel, a self-assembling viral nanoparticle platform (Nov-S-Catcher003) was generated by introducing SpC into the norovirus S domain (Nov-S). Nov-S-Catcher003 was expressed in *E. coli* and self-assembled into uniform nanoparticles, and conjugation of SpT-tagged TC-1 and B16F10 antigens induced effective antitumor immune responses, supporting the feasibility of modular anti-tumor vaccination using self-assembling viral nanoparticles [175].

Additional bacterial platforms have been used to promote myeloid activation and trained immunity. BG/OVA@EcN was produced by coating inactivated *E. coli* Nissle 1917 with PEI and then incorporating OVA and the immunostimulant β -glucan. BG/OVA@EcN recruited monocytes/macrophages and DCs at the injection site and exhibited enhanced uptake, consistent with β -glucan-mediated recognition. Trained monocytes/macrophages subsequently differentiated toward an M1-like phenotype within tumors, promoting tumor-specific immunity and long-term immune memory [176].

E. coli OMVs engineered to express a polybia-mastoparan O (MPI) fusion protein reduced the viability of bladder cancer cell lines (MB49 and UMUC3). The immunostimulatory components of Gram-negative OMVs promoted antigen-presenting cell activation and CTL infiltration, with increased secretion of antitumor cytokines, including IL-6, TNF- α , and IFN- γ [177].

Finally, OMV-SIRP α @CaP/GM-CSF was generated by harvesting OMVs from *E. coli* expressing SIRP α -Fc, biomimeticizing the OMVs with calcium phosphate (CaP), and loading GM-CSF. This formulation incorporates PAMPs, including OMV-derived LPS, and can induce trained immunity in macrophages. OMV-SIRP α @CaP/GM-CSF suppressed tumor growth in both MC38 (tumor-associated macrophage (TAM)-high, T cell-low) and B16F10 (TAM-low, T cell-high) models. In MC38, antitumor activity reflected contributions from trained immunity and T cell responses, whereas in B16F10, efficacy was attributed primarily to trained immunity [178].

Overall, chemical modification, genetic engineering, and related molecular tailoring strategies enable precise optimization of CNP-based cancer nanovaccines by enhancing targeting, antigen presentation, and immunostimulatory signaling. These

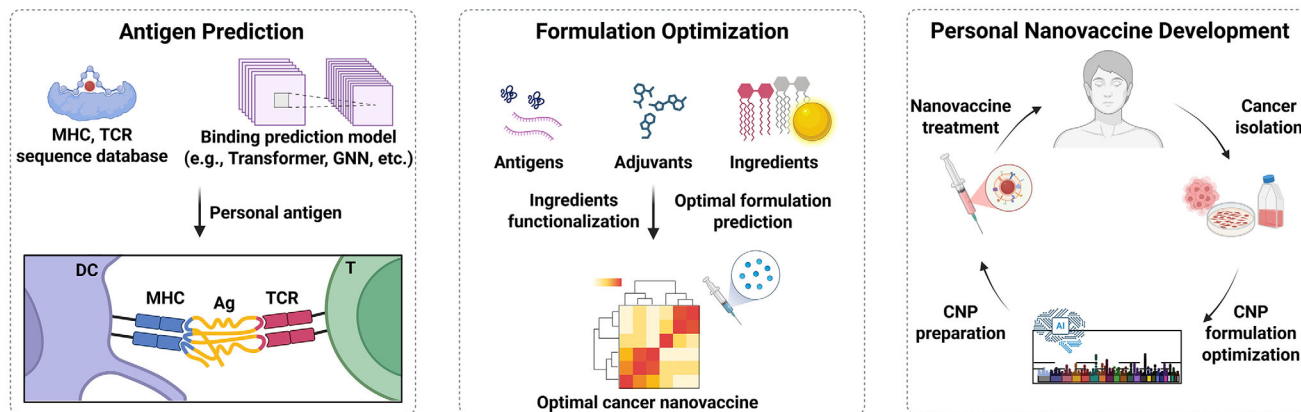


FIGURE 8 | Perspectives on the application of cancer nanovaccines and AI. In cancer nanovaccine research, integration of AI architectures (e.g., Transformer, GNN, and Attention architectures) can be broadly grouped into three areas: tumor antigen prediction, nanovaccine formulation and component optimization, and development of personalized cancer nanovaccines. AI-enabled antigen prediction can leverage existing antigen databases to support identification of candidate antigen sequences and to assess properties relevant to immunogenicity, including peptide structure, MHC loading, and potential TCR engagement. AI can also accelerate formulation development by optimizing functionalization strategies and component ratios. Personalized cancer nanovaccines remain challenging due to high costs and difficulties in achieving manufacturing reproducibility and batch-to-batch uniformity. Although patient-derived tumor cells harbor diverse antigen repertoires, vaccine development using patient materials is limited by challenges in cell culture, formulation optimization, production consistency, and clinical evaluation. AI is expected to contribute to process optimization for patient-derived manufacturing workflows and to the development of standardized criteria aligned with regulatory requirements. Created in BioRender. Noh, I. (2025) <https://BioRender.com/gb97th9>.

approaches preserve cancer antigens while enabling coordinated activation of DCs and T cells, induction of trained immunity, and other multimodal antitumor responses, highlighting the therapeutic potential of functionalized cell membrane-derived nanoparticles in cancer immunotherapy.

Despite its improved anti-cancer immune responses, several challenges remain for clinical uses. First, when patient-derived cancer cells are used, they must be cultured, which requires prior optimization of culture conditions. In addition, large-scale cell culture is necessary to obtain sufficient cancer cell membrane for vaccination. Furthermore, even in engineered CNPs, current studies are limited to a few specific approaches. In most cases, additional adjuvant is still required, suggesting that CNP technology should focus on the direct functionalization of membrane.

4 | Challenges and Prospects

We propose that successful development of anticancer nanovaccines will require parallel advances in (i) personalized antigen sequence prediction and (ii) delivery vehicle engineering (Figure 8).

4.1 | Personalized Antigen Prediction and Safety-First Mechanism

The identification and prioritization of antigen sequences remain major bottlenecks in cancer vaccine development [5, 179]. Although patient tumors harbor numerous somatic mutations, only a small fraction of the derived antigens are sufficiently

immunogenic to elicit robust T cell responses. The effective activation of antigen-specific T cells depends on multiple processing steps, including proteasomal cleavage, transporter associated with antigen processing (TAP)-mediated transportation, peptide loading onto MHC molecules, and recognition by TCRs [15] (Figure 1).

While AI antigen prediction models are advancing, significant limitations remain in their practical clinical application. Most existing models are trained on large-scale databases, which are often biased toward specific cancer types and common HLA alleles. Consequently, research on AI-based personalized cancer vaccine sequences remains in its early stages of development. The AI antigen design prediction tool requires the integration of advanced transcriptomic analysis and processing technology. For instance, certain patients fail to respond to ICBs, indicating that TMEs and immune systems vary significantly among individuals. Currently, various models are being developed to predict responsiveness to ICBs and associated immune-related adverse effects (irAEs) [180, 181]. Therefore, personalized cancer vaccine models are needed to include antigen prediction and analyze the correlations between the tumor, TME, and immune cells.

Furthermore, AI-generated antigens have the risk of inducing off-target effects by acting on normal cells, which remains an important challenge for clinical translation. To address this, a filtering mechanism should be implemented during antigen design, referencing a comprehensive human self-antigen database to exclude sequences with predicted cross-reactivity. Future AI frameworks must move beyond merely identifying sequences that bind to tumor antigens; it is essential to establish a “safety-first” system that prioritizes immunologically safe antigens.

4.2 | Clinical Translation of AI-Designed Nanoparticle and CNP Platforms

Research on AI-based nanocarrier development is rapidly advancing. However, regulatory and temporal barriers remain for practical clinical translation. This issue is a critical bottleneck in the development of novel lipid and polymer components, where Transformer models and GNNs are primarily utilized. Newly synthesized materials require several years for preclinical and clinical evaluation, making it difficult to provide immediate administration to patients with urgent needs. Therefore, to meet immediate clinical demands, it is essential to develop AI platforms for patient-specific formulation optimization in parallel with research on de novo compound synthesis. Such platforms should focus on reconfiguring approved drugs and carriers into optimal ratios based on individual patient characteristics, including age, weight, and sex. Furthermore, predictive models for the in vivo behavior, toxicity, and non-specific immune responses of nanoparticles must also be developed.

CNPs offer several advantages for cancer nanovaccine development, as they provide biomimetic properties derived from the cell membrane of the source. However, the clinical translation of CNPs remains constrained by manufacturing challenges. Personalized CNP-based vaccines require standardized membrane extraction and purification workflows, reproducible production, and well-defined quality attributes.

AI-enabled process optimization and real-time quality monitoring may help address these limitations and support patient-specific design decisions aligned with regulatory expectations. AI could further expand the utility of CNP platforms by enabling the systematic optimization of key design variables, including the selection of membrane sources, auxiliary lipid composition, membrane-to-core ratios, and surface functionalization strategies.

5 | Summary

In this review, we summarize the key developments in cancer vaccines and cancer nanovaccines. Following the identification of tumor antigens, extensive efforts have been focused on eliciting antitumor immunity using antigen-based vaccination strategies. Early protein-based vaccines were followed by the emergence of mRNA vaccine technologies, and subsequent studies prioritized improving delivery efficiency through nanoparticle formulations. Although LNP-formulated mRNA nanovaccines currently predominate, recent studies have increasingly emphasized CNP platforms to leverage native antigen repertoires and mitigate the limitations associated with fully synthetic carriers.

Personalized therapy is a growing focus in contemporary healthcare. Given the heterogeneity of tumor antigen expression, personalized cancer nanovaccines derived from the cell membranes of cancer cells are positioned as a promising approach within the broader landscape of precision therapeutics.

The integration of AI with CNP technology holds the potential to reshape next-generation cancer vaccine development. AI can support multiple stages of the pipeline, including antigen and

epitope identification, formulation and design optimization, and prediction of clinical responses. In parallel, CNPs offer a natural biological interface that can improve biocompatibility and enable multivalent antigen presentation, addressing challenges often unsolved by synthetic nanoparticles alone. Future research must prioritize clinical validation of AI-guided CNP platforms and establish scalable, reproducible manufacturing strategies compatible with regulatory translation.

Author Contributions

All authors have written this review paper with contributions.

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Conflicts of Interest

The authors declare no conflict of interest.

Data Availability Statement

Data sharing not applicable to this article as no datasets were generated or analyzed during the current study.

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